Basic Of Internet

Lec 3

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Decoding IP Address

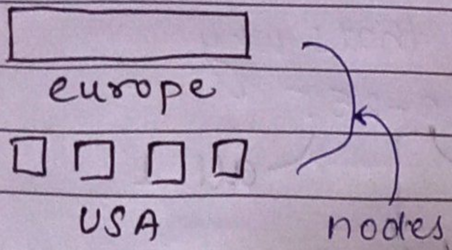
→ IP Address

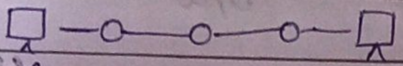
1974

↳ TCP/IP => ARPANET

↓

protocol (rules)





in b/w get confused
to follow both systems differently

TCP/IP →

Standard protocols

source
& destination address (inside packets)

IP :- Internet protocol

same as post office system,
needs address of where to send packet.

↓

IP address (unique address)

32-bit

IPv4 – 32 bit (8 x 4)

4 bytes ✓

1 byte = 8 bits

192 . 36 . 15 . 8 (format)

↓ ↓ ↓ ↓

0-255 0-255 0-255 0-255 (range)

whatsmyip.org
Rohit Negi : 122.162.151.31

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2^5
⇓
32 bit = $\boxed{8 \text{ bit}}$. $\boxed{8 \text{ bit}}$. $\boxed{8 \text{ bit}}$. $\boxed{8 \text{ bit}}$

$2^8 = 256$

↓

that's why

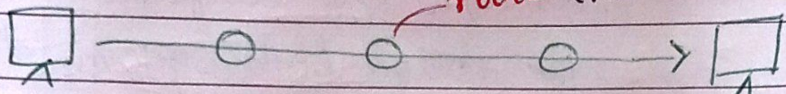
0 – 255

all 0 — all 1

↓

b/w

router (forward the msg)



192.36.58.24 — HELLO — 102.52.19.25

Source address — destination address

When device connected to internet they get IP address.

Max IP address we can give to devices accross world

$2^5 \to 2^{32}$ = 400 crores (approx)

but devices all over world connected to internet must be above 1000 crore +

to consider for future

$2^6 \times$

IPv6

$2^7 \Rightarrow 128$ bits — 2^{128} devices can

(hexa decimal form)

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Who allocate IP address? Internet Service Provider (ISP)
jio, airtel, etc.

How we see google.com without IP address?

user client —request→ Server
←respond—

Server — serves request of client

Domain Name System (DNS) - 1980
└ familiar with letters

request - comment, dlt

DNS= Domain Name $\xrightarrow{\text{convert}}$ IP Address

User —Youtube→ ISP —cache (temp memory)

192.36.12.59
(YT.com)

If ISP does'nt have IP address of user request?

ISP —request IP address→ .com .in .us — yt.com, google.com, ...com

Root Name Server

user ← ISP ← if in cache — IP address

All Working

User —request YT.com→ ISP → .com .in .net → .com → Youtube .com

User | ISP (cache) | Root Name Server | Top Level Domain | Authoritative Name Server └ IP add.

if not in cache

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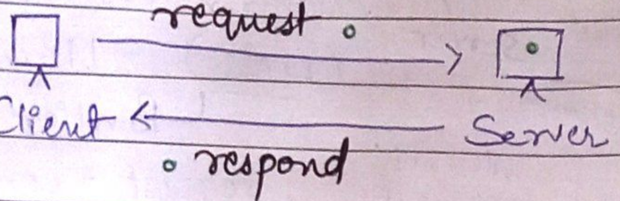
Static IP address → a fixed IP add. for a device

Dynamic IP address → IP address changes



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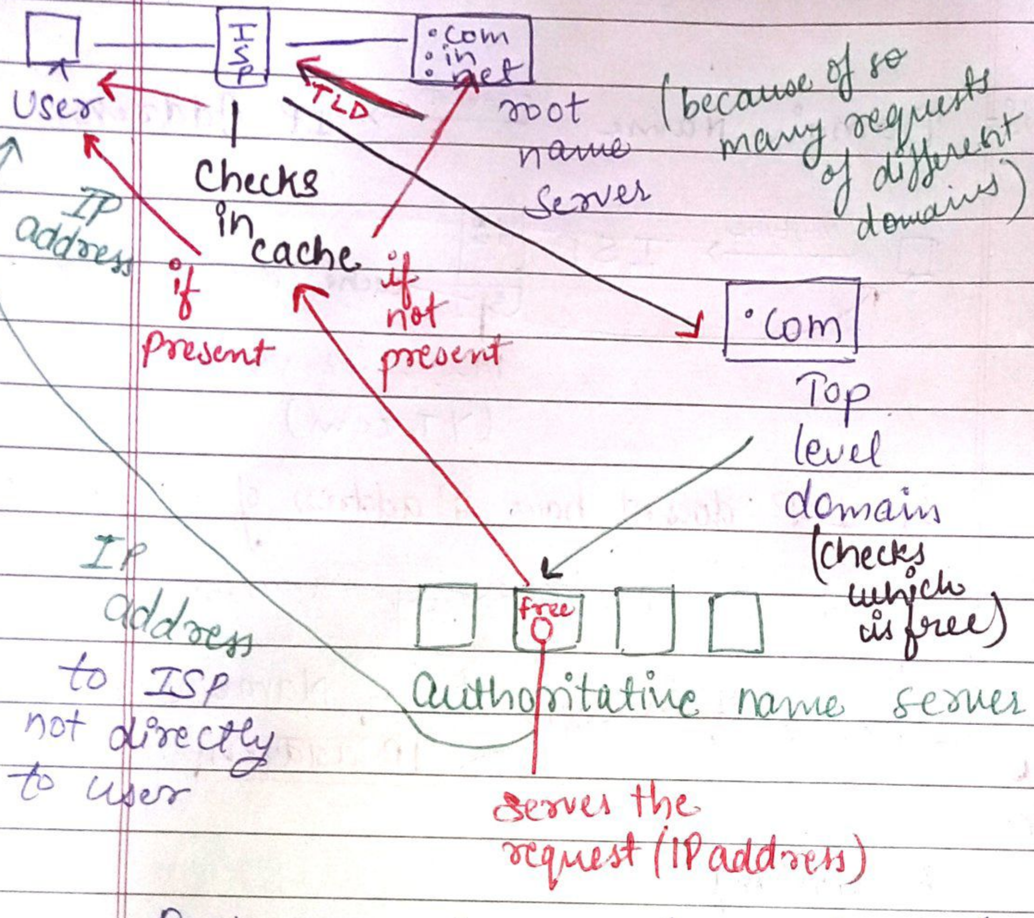

Root name Server – only 13 in world

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Classes in IP Addresses

① Class A 60 % of IP address (128)

Total IP addresses 2^{32} = 256
So 256 / 2 = 128 0-255 . 0-255 . 0-255 . 0-255
50%

0-127 . any . ___ . ___

② Class B 25 % of IP address (64)

128-191 . ___ . ___ . ___

③ Class C 12.5 % of IP address (32)

192-223 . ___ . ___ . ___

④ Class D 6.25% of IP address (16)

224-239 . ___ . ___ . ___

⑤ Class E 6.25 % of IP address (16)

240-255 . ___ . ___ . ___